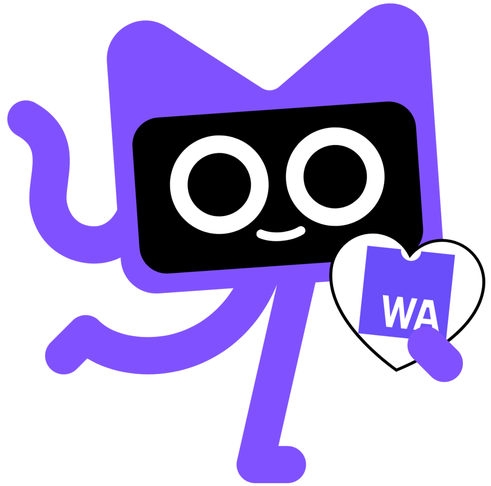
JETBRAINS



Building Web Apps with Kotlin and Wasm*

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* kotl.in/wasm

DisclAImer

Kotlin

by JETBRAINS

* kotlinlang.org

Multiplatform

Android iOS Web Desktop Server

* kotlinlang.org/multiplatform

Multiplatform

Android iOS Web Desktop Server

Kotlin/JVM Kotlin/Native Kotlin/JS Kotlin/Wasm

* kotlinlang.org/multiplatform

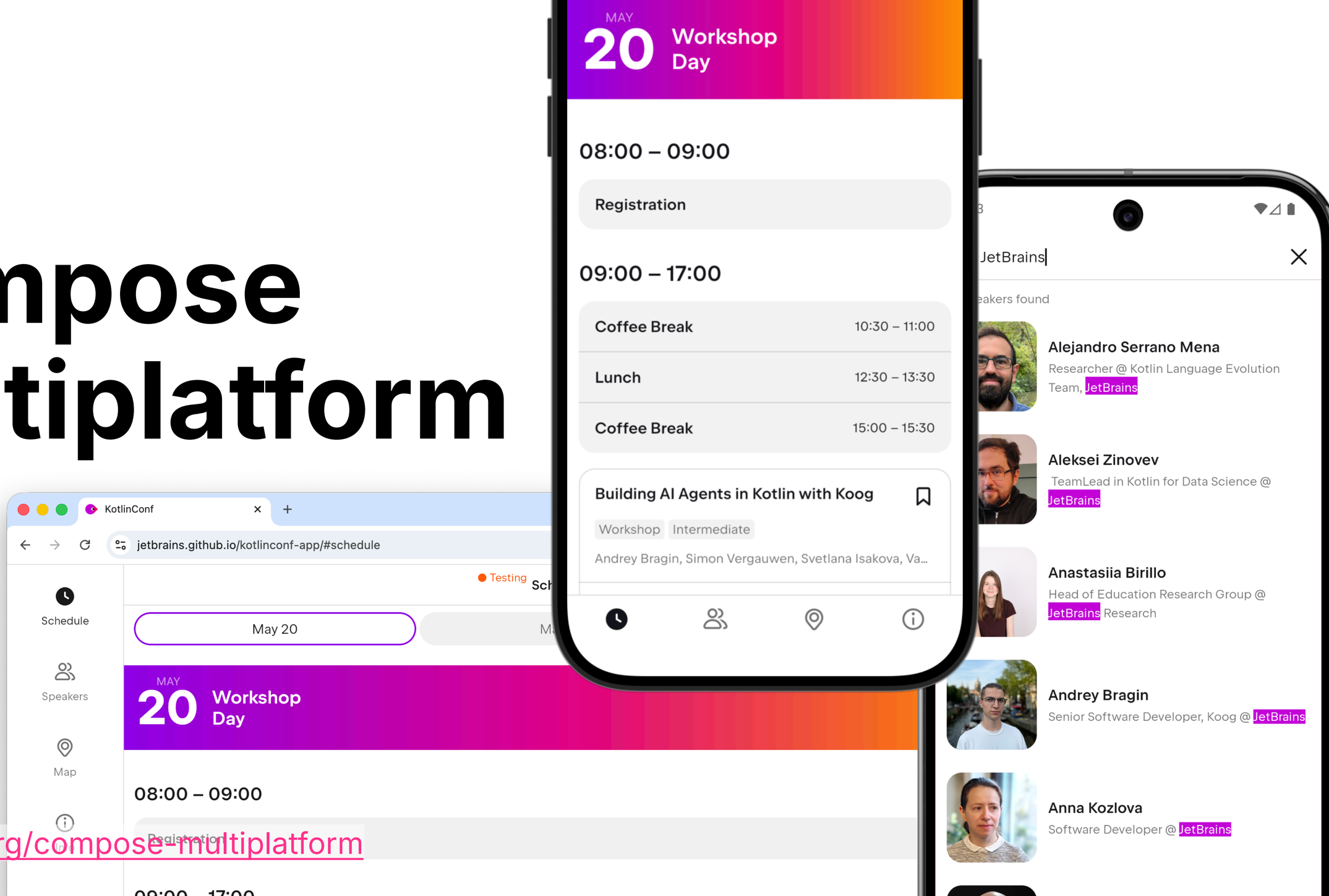
Building Web Apps

Directly work with DOM / Browser APIs

Demo

* github.com/Kotlin/kotlin-wasm-browser-template
* Debug it in DevTools 👉 wasmdemo.github.io

Compose Multiplatform



* kotlinlang.org/compose-multiplatform

Demo

* github.com/Kotlin/kotlin-wasm-compose-template

Compose HTML

Demo

Overview

	Browser API / Direct DOM manip.		
Size of distribution	++		
Performance	!+		
Declarative	x		
Maintainability	x		
Shared UI Code	x		

Overview

	Browser API / Direct DOM manip.	Compose web (Canvas-based)
Size of distribution	++	x
Performance	!+	++
Declarative	x	+
Maintainability	x	++
Shared UI Code	x	+

Overview

	Browser API / Direct DOM manip.	Compose web (Canvas-based)	Compose HTML (DOM-based)
Size of distribution	++	x	+
Performance	!+	++	+
Declarative	x	+	+
Maintainability	x	++	?+
Shared UI Code	x	+	x

Kotlin

by JETBRAINS

* kotlinlang.org

The Best UX for end users

The Best DX for developers

Current and Future Works

Areas

Active exploration

Compilation Hints

Branch Hints

SIMD

Multithreading [wa][kt]

ESM Integration

Component Model [wa][kt]

JSPI

Interop with JS / Browser APIs

Memory Control

Multi-module output [kt]

Stack Switching [wa][kt]

Compact imports

Type Imports

Multibyte Array Access

Areas

Future Work

Compilation Hints [wa][kt]

Branch Hints [wa][kt]

SIMD

Multithreading

ESM Integration [wa][kt]

Component Model

JSPI

Interop with JS / Browser APIs [kt]

Memory Control

Multi-module output

Stack Switching

Compact imports

Type Imports

Multibyte Array Access

Areas

Under consideration (Follow)

Compilation Hints

Branch Hints

SIMD [wa]

Multithreading

ESM Integration

Component Model

JSPI [wa]

Interop with JS / Browser APIs

Memory Control [wa]

Multi-module output

Stack Switching

Compact imports [wa]

Type Imports [wa]

Multibyte Array Access [wa]

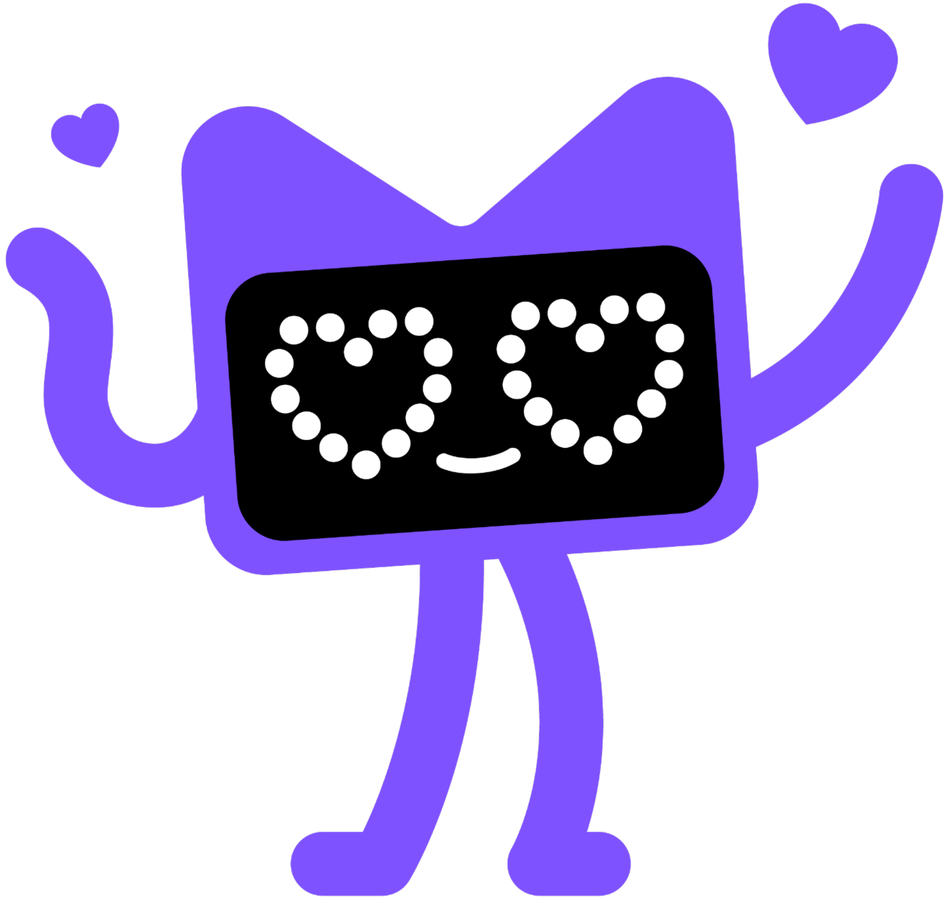
Here’s what we covered:

- Ways to build UI for Web apps
- Tooling, IDE, Debugger
- Size of different applications
- Current and future works

kotl.in/wasm bashorov.com

Thank you!

Zalim Bashorov



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Thank you!

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